

Vector Calculus

1 Euclidean space

1.1 Basic Concepts

We define the **Euclidean space** $\mathbb{R}^n$, $n \geq 1$, as the set of all elements in $\mathbb{R}$:

$$x = (x_1, x_2, \dots, x_n) \quad \text{where } x_i \in \mathbb{R} \text{ for } i = 1, 2, \dots, n.$$

The **Standard basis** of $\mathbb{R}^n$ is given by the vectors:

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, \dots, 0), \quad \dots, \quad e_n = (0, 0, \dots, 1).$$

Then, any vector $x \in \mathbb{R}^n$ can be expressed as a linear combination of the standard basis vectors:

$$x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n.$$

1.2 Properties of Euclidean Space

The Euclidean space $\mathbb{R}^n$ has the following properties:

- **Addition:** For any $x, y \in \mathbb{R}^n$, the sum $x + y$ is defined component-wise:

$$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n).$$

- **Scalar Multiplication:** For any scalar $\alpha \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$, the product αx is defined as:

$$\alpha x = (\alpha x_1, \alpha x_2, \dots, \alpha x_n).$$

- **Scalar Distributivity:** For any scalars $\alpha, \beta \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$:

$$(\alpha + \beta)x = \alpha x + \beta x.$$

- **Vector Distributivity:** For any scalar $\alpha \in \mathbb{R}$ and vectors $x, y \in \mathbb{R}^n$:

$$\alpha(x + y) = \alpha x + \alpha y.$$

- **Scalar Commutativity:** For any scalars $\alpha, \beta \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$:

$$\alpha(\beta x) = (\alpha\beta)x.$$

- **Multiplicative Identity:** For any vector $x \in \mathbb{R}^n$:

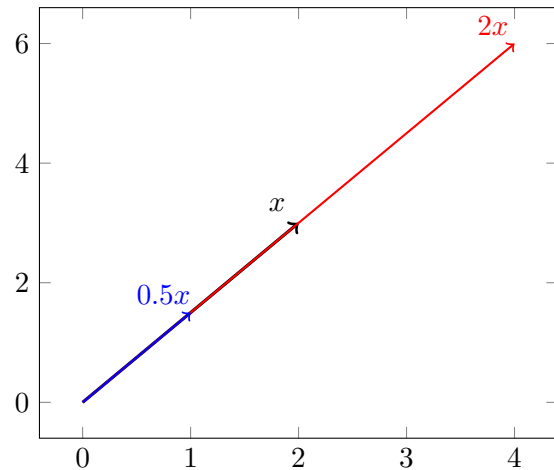
$$1 \cdot x = x.$$

- **Multiplicative Zero:** For any vector $x \in \mathbb{R}^n$:

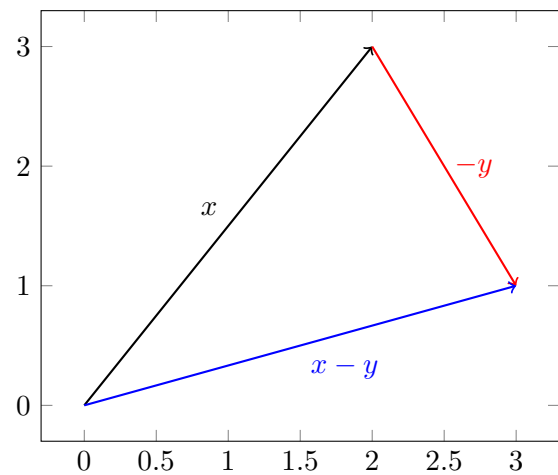
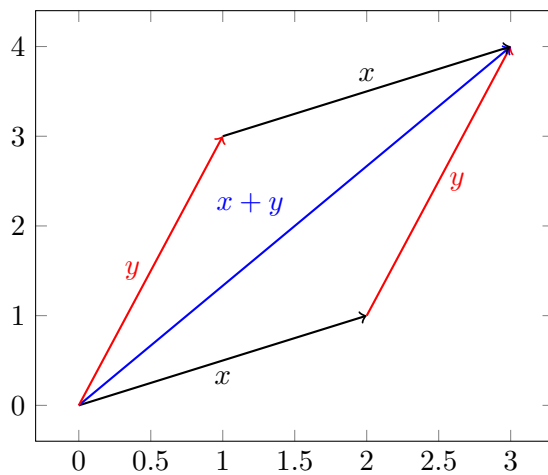
$$0 \cdot x = 0.$$

1.3 Geometric Interpretation

The Euclidean space $\mathbb{R}^n$ can be visualized as an n -dimensional space where each point corresponds to a vector. Given a vector $x = (x_1, x_2, \dots, x_n)$, and a scalar $\alpha \in \mathbb{R}$, the vector αx represents a point that is scaled by the factor α in the direction of x :



Given two vectors $x, y \in \mathbb{R}^n$, their sum $x + y$ represents the vector obtained by placing the tail of y at the head of x . Similarly, the difference $x - y$ can be visualized by reversing the direction of y and adding it to x .



1.4 Norm and Distance

The Euclidean space is a **normed vector space** with the norm defined as:

$$\begin{aligned} \|\cdot\| : \mathbb{R}^n &\rightarrow \mathbb{R} \\ x &\mapsto \|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}. \end{aligned}$$

The norm satisfies the following properties for all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- **Non-negativity:** $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$.
- **Scalar Multiplication:** $\|\alpha x\| = |\alpha| \|x\|$.
- **Triangle Inequality:** $\|x + y\| \leq \|x\| + \|y\|$.

Remark

Associated with the norm is the **distance function** $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by:

$$d(x, y) = \|x - y\|.$$

The distance function satisfies the following properties for all $x, y, z \in \mathbb{R}^n$:

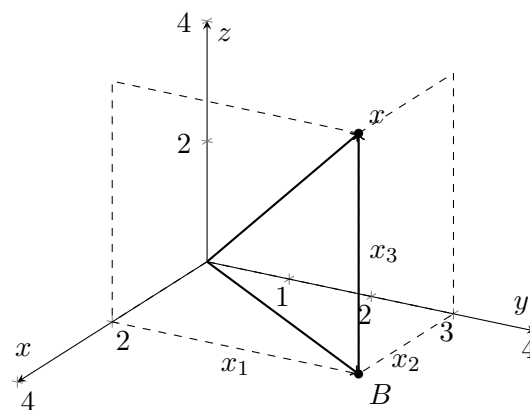
- **Non-negativity:** $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$.
- **Symmetry:** $d(x, y) = d(y, x)$.
- **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

1.4.1 Norms in $\mathbb{R}^3$

In $\mathbb{R}^3$, the norm of a vector $x = (x_1, x_2, x_3)$ is given by:

$$\|x\| = d(O, x) = \sqrt{x_1^2 + x_2^2 + x_3^2},$$

where O is the origin and B is the projection of x onto the xy -plane:



1.5 Inner or Scalar Product

Let $x, y \in \mathbb{R}^n$. The **inner product** (or **scalar product**) of x and y is defined as:

$$x \cdot y = \langle x, y \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$$

The inner product satisfies the following properties for all $x, y, z \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- **Positive Definite:** $x \cdot x \geq 0$ and $x \cdot x = 0$ if and only if $x = 0$.
- **Symmetry:** $x \cdot y = y \cdot x$.
- **Bilinear Form:** $\langle \lambda x + \mu y, z \rangle = \lambda \langle x, z \rangle + \mu \langle y, z \rangle$ for all $\lambda, \mu \in \mathbb{R}$.

1.6 Cauchy-Schwarz Inequality

For any vectors $x, y \in \mathbb{R}^n$, the Cauchy-Schwarz inequality states that:

$$|x \cdot y| \leq \|x\| \|y\|.$$

Equality holds if and only if x and y are linearly dependent.

Proof. If $y = \lambda x$ for some $\lambda \in \mathbb{R}$, then:

$$|x \cdot y| = |x \cdot (\lambda x)| = |\lambda| \|x\|^2 = \|x\| \|\lambda x\|.$$

If x and y are linearly independent, consider the vector $z = \lambda x + y$ for some $\lambda \in \mathbb{R}$. Then:

$$0 \leq \langle z, z \rangle = \|z\|^2 = \langle \lambda x + y, \lambda x + y \rangle.$$

Expanding this, we have:

$$\|z\|^2 = \lambda^2 \|x\|^2 + 2\lambda \langle x, y \rangle + \|y\|^2,$$

which is a quadratic function in λ . For $f(\lambda) \geq 0$ for all λ , the discriminant must be non-positive:

$$(2 \langle x, y \rangle)^2 - 4 \|x\|^2 \|y\|^2 \leq 0.$$

Simplifying, we get:

$$4 \langle x, y \rangle^2 \leq 4 \|x\|^2 \|y\|^2,$$

which leads to:

$$|x \cdot y| \leq \|x\| \|y\|.$$

□

1.7 Theorem: Inner Product and Norm

For any vectors $x, y \in \mathbb{R}^n$, the following relation holds:

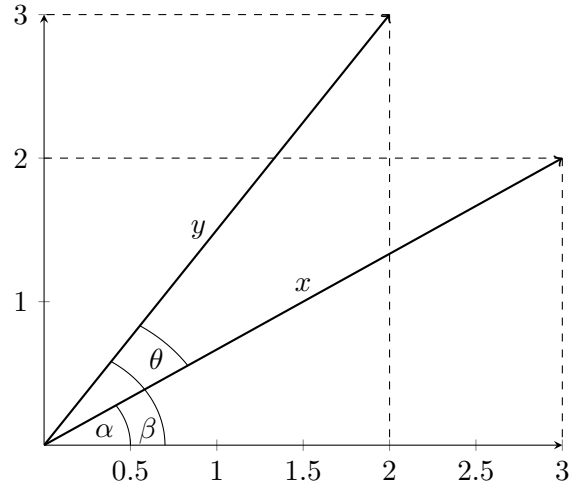
$$\langle x, y \rangle = \|x\| \|y\| \cos \theta,$$

where θ is the angle between the vectors x and y .

Proof. Consider the vectors x and y in $\mathbb{R}^2$. Let the angle between x and the positive x -axis be α , and the angle between y and the positive x -axis be β . Then, the angle between x and y is given by $\theta = \beta - \alpha$. We have:

$$x_1 = \|x\| \cos \alpha, \quad x_2 = \|x\| \sin \alpha,$$

$$y_1 = \|y\| \cos \beta, \quad y_2 = \|y\| \sin \beta.$$



Therefore, the inner product is:

$$\begin{aligned} \langle x, y \rangle &= x_1 y_1 + x_2 y_2 \\ &= \|x\| \cos \alpha \cdot \|y\| \cos \beta + \|x\| \sin \alpha \cdot \|y\| \sin \beta \\ &= \|x\| \|y\| (\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ &= \|x\| \|y\| \cos(\beta - \alpha) \\ &= \|x\| \|y\| \cos \theta. \end{aligned}$$

□

Remark

From the theorem, we can derive the formula for the angle θ between two vectors x and y :

$$\theta = \arccos \left(\frac{\langle x, y \rangle}{\|x\| \|y\|} \right).$$

Note that if $x \perp y$ (i.e., x is perpendicular to y), then $\langle x, y \rangle = 0$ and thus $\theta = \frac{\pi}{2}$.

Examples

$C(a, b)$ is the set of continuous functions on the interval (a, b) . For any $f, g \in C(a, b)$, we define the inner product as:

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx.$$

We can add a weighted inner product as follows:

$$\langle f, g \rangle_w = \int_a^b f(x)g(x)w(x) dx,$$

where $w(x)$ is a weight function such as $w(x) = e^{-x}$ or $w(x) = 1/x^2$.

We might as well use the family of **orthogonal functions**:

$$\{\cos nx, \sin nx \mid n = 0, 1, 2, \dots\} \quad \text{Fourier family.}$$

$$0 = \int_0^{2\pi} \cos nx \sin nx dx = \int_0^{2\pi} \cos nx \cos mx dx = \int_0^{2\pi} \sin nx \sin mx dx, \quad n \neq m.$$

1.8 Vector Product in $\mathbb{R}^3$

Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be two vectors in $\mathbb{R}^3$. The **vector product** (or cross product) of x and y is defined as:

$$x \times y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \mathbf{k},$$

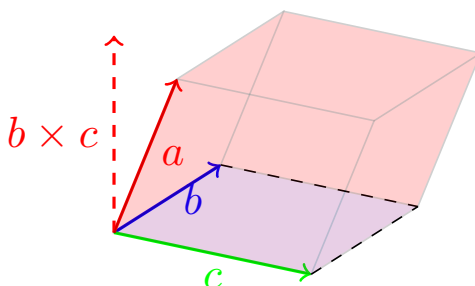
where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit vectors in $\mathbb{R}^3$.

1.8.1 Geometric Interpretation

To visualize the vector product, consider the triple product $a \cdot (b \times c)$, which represents the volume of the parallelepiped formed by the vectors a, b , and c :

$$a \cdot (b \times c) = (a_1, a_2, a_3) \cdot \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix} \mathbf{k} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

If $a \in \text{span}\{b, c\}$, then the volume is zero, indicating that the vectors are coplanar.



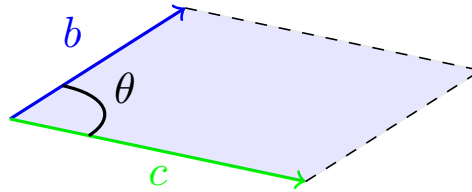
Also,

$$\begin{aligned}\|b \times c\|^2 &= \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}^2 = (b_2c_3 - b_3c_2)^2 + (b_1c_3 - b_3c_1)^2 + (b_1c_2 - b_2c_1)^2 = \\ &= (b_1^2 + b_2^2 + b_3^2)(c_1^2 + c_2^2 + c_3^2) - (b_1c_1 + b_2c_2 + b_3c_3)^2 = \|b\|^2\|c\|^2 - (b \cdot c)^2 = \|b\|^2\|c\|^2 \sin^2 \theta.\end{aligned}$$

Thus, we have:

$$\|b \times c\| = \|b\|\|c\| \sin \theta,$$

where θ is the angle between vectors b and c . This represents the area of the parallelogram formed by b and c .



2 Level Curves

Let $A \subset \mathbb{R}^n$ be a subset of $\mathbb{R}^n$ and let $f : A \rightarrow \mathbb{R}$ be a real-valued function defined on A . For any constant $c \in \mathbb{R}$, the **level set** of f corresponding to the value c is defined as:

$$L_c = \{x \in A \mid f(x) = c\}.$$

In the case where $n = 2$, the level sets are called **level curves** or **contour lines**. In the case where $n = 3$, the level sets are called **level surfaces**.

Example (Chapter 12, Ex 3)

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = xy.$$

The level curves of f for different values of c are given by:

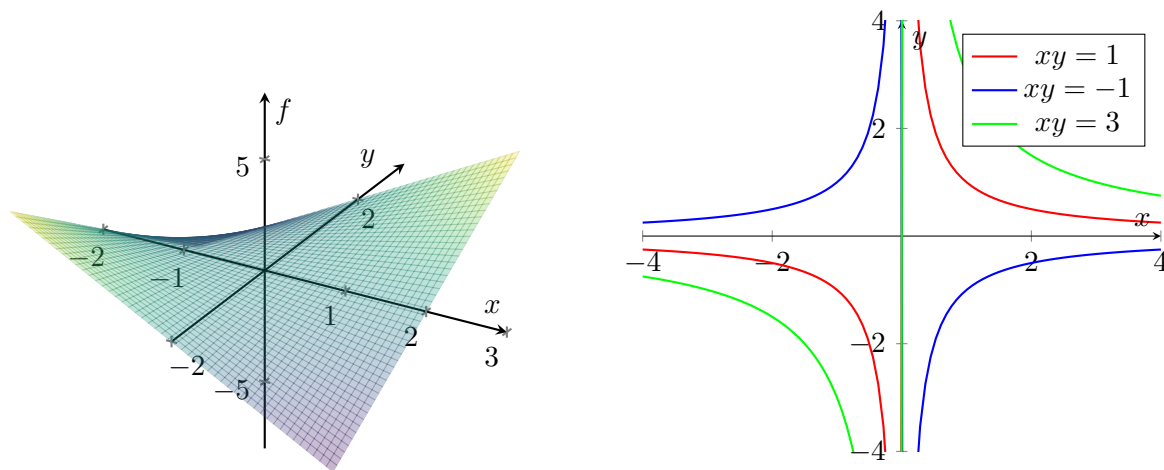
- For $c = -1$: $L_{-1} = \{(x, y) \in \mathbb{R}^2 \mid xy = -1\}$.
- For $c = 1$: $L_1 = \{(x, y) \in \mathbb{R}^2 \mid xy = 1\}$.
- For $c = 3$: $L_3 = \{(x, y) \in \mathbb{R}^2 \mid xy = 3\}$.

2.1 Graph of a Function

The **graph** of a function $f : A \rightarrow \mathbb{R}$ is defined as the set of points in $\mathbb{R}^{n+1}$ given by:

$$G_f = \{(x, f(x)) \mid x \in A\}.$$

If for instance, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, then the graph of f is a surface in $\mathbb{R}^3$.



Remark

Level sets $F : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$; U open set:

$$F^{-1}(c) = \{x \in U \mid F(x) = c\}, \quad c = (c_1, c_2, \dots, c_m)$$

In terms of the coordinate functions $F = (f_1, f_2, \dots, f_m)$, we have:

$$F(x)(c), \text{ for } x \in \mathbb{R}^n \implies f_i(x_1, x_2, \dots, x_n) = c_i, \quad i = 1, 2, \dots, m.$$

Hence,

$$F^{-1}(c) = \bigcap_{i=1}^m \{x \in \mathbb{R}^n \mid f_i(x) = c_i\}.$$

Example

Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by:

$$F(x, y, z) = (x^2 + y^2 + z^2 - 1, 2x^2 + 2y^2 - 1).$$

The coordinate functions are:

$$f_1^{-1}(0) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 - 1 = 0\},$$

$$f_2^{-1}(0) = \{(x, y, z) \in \mathbb{R}^3 \mid 2x^2 + 2y^2 - 1 = 0\}.$$

To find the level set we solve the system of equations:

$$x^2 + y^2 + z^2 - 1 = 0,$$

$$2x^2 + 2y^2 - 1 = 0.$$

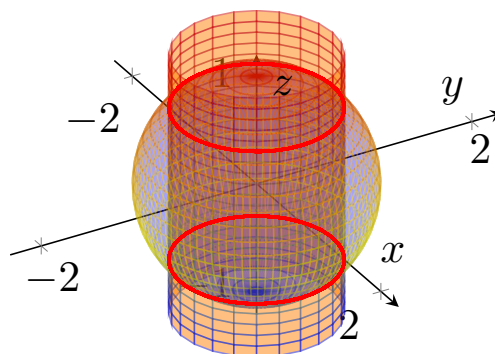
From the second equation, we have:

$$x^2 + y^2 = \frac{1}{2}.$$

Substituting this into the first equation gives:

$$\frac{1}{2} + z^2 - 1 = 0 \implies z^2 = \frac{1}{2} \implies z = \pm \frac{1}{\sqrt{2}}.$$

Thus, the level set $F^{-1}(0)$ is the intersection of the sphere $x^2 + y^2 + z^2 = 1$ and the cylinder $2x^2 + 2y^2 = 1$.



Remarks

- There is a connection between level sets and graphs. Firstly, the graph of a function $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ can be viewed as the level set of the function $F : A \times \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$F : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad F(x, z) = f(x) - z.$$

- The level curves allow us to draw tri-dimensional surfaces on a two-dimensional plane.
- The distance between two level curves (they can never intersect) gives us information about the steepness of the surface. If the level curves are close together, the surface is steep; if they are far apart, the surface is flat.

3 Limits of Functions

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function, and let $a \in \mathbb{R}^n$ be a point in its domain. We say that the limit of $f(x)$ as x approaches a is $L \in \mathbb{R}^m$, denoted by:

$$\lim_{x \rightarrow a} f(x) = L,$$

if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in \mathbb{R}^n$ satisfying $0 < \|x - a\| < \delta$, we have:

$$\|f(x) - L\| < \epsilon.$$

3.1 Methods to Compute Limits

1. **Applying the Definition:** We must prove that for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$0 < (x, y) - (a, b) < \delta \implies |f(x, y) - L| < \epsilon.$$

2. **Iterative limits:**

$$\lim_{x \rightarrow a} \left(\lim_{y \rightarrow b} f(x, y) \right) \neq \lim_{y \rightarrow b} \left(\lim_{x \rightarrow a} f(x, y) \right).$$

This gives non-existence of the limit, so we cannot prove existence by this method.

3. **Following families of functions:** If we can find two different paths approaching the same point that yield different limit values, then the limit does not exist.